



Blood Donation Management System (BDMS)

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Our Team



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**Overall
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OVERALL DESCRIPTION

PRODUCT PERSPECTIVE

Currently, blood banks in Egypt are either independent or limited to specific hospital networks. This leads to delays in finding donors, especially for rare blood types, and forces hospitals to ask patients' families to donate or pay for blood. The proposed system provides a unified platform that connects donors directly with nearby hospitals in real time, making the process faster, safer, and more accessible.

In Egypt, the current blood donation rate is only 1 donor per 1,000 citizens, while the annual national demand exceeds 1.6 million blood bags. This shortage creates a critical gap between supply and demand, especially during emergencies and for rare blood types—making an intelligent blood management system essential.

source : [المصري اليوم](#)



USERS CLASSES



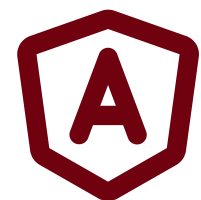
DONOR MODULE:

ANY SUITABLE DONOR



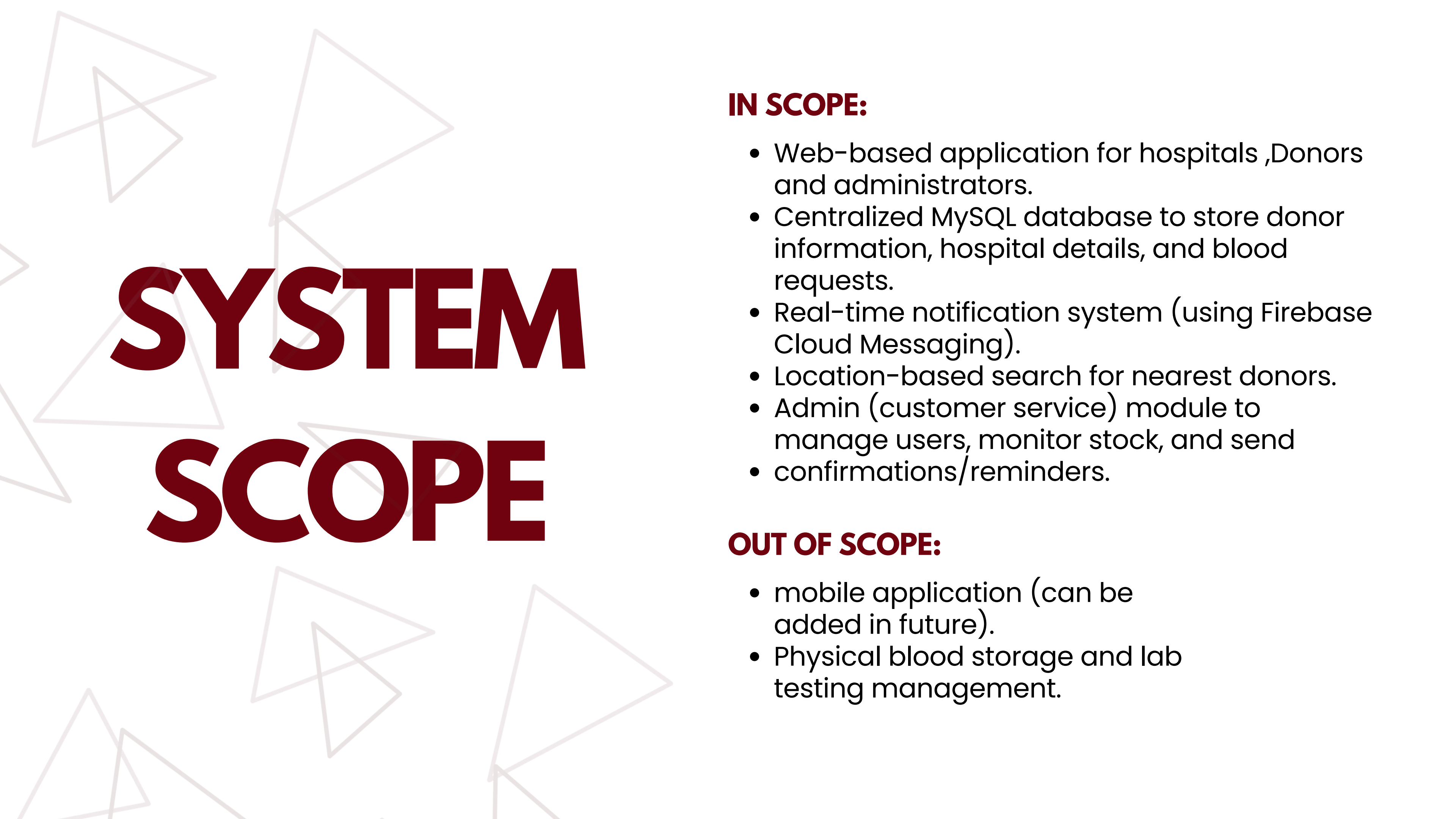
HOSPITAL MODULE:

BLOOD BANK NURSE



ADMIN MODULE:

COMPANY ENGINEER



SYSTEM SCOPE

IN SCOPE:

- Web-based application for hospitals ,Donors and administrators.
- Centralized MySQL database to store donor information, hospital details, and blood requests.
- Real-time notification system (using Firebase Cloud Messaging).
- Location-based search for nearest donors.
- Admin (customer service) module to manage users, monitor stock, and send
- confirmations/reminders.

OUT OF SCOPE:

- mobile application (can be added in future).
- Physical blood storage and lab testing management.

OPERATING ENVIRONMENT

- WEB:
 - frontEnd(React.js,Tailwind)
 - BackEnd(Node.js)
- DATABASE: MYSQL
- AUTHENTICATION: FIREBASE AUTHENTICATION



ASSUMPTIONS AND DEPENDENCIES

- USERS HAVE INTERNET ACCESS.
- HOSPITALS ARE PRE-APPROVED BY ADMIN.
- DONORS PROVIDE ACCURATE LOCATION AND CONTACT INFORMATION.
- GOOGLE MAPS/GEOCODING API (OPTIONAL FOR BETTER LOCATION ACCURACY).

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

FUNCTIONAL REQUIREMENTS



DONOR MODULE

IDs	Functional Requirement
FR-D1	Donor shall be able to register with name, age, phone, address, blood type, gender, and location, donation count, weight, Hight, last donation date, diagnosis, image.
FR-D2	Donor shall login using username and password.
FR-D3	Donor shall receive push notification when a nearby hospital needs his/her blood type.
FR-D4	Donor shall be able to Accept or Reject a blood donation request.
FR-D5	Donor shall view details of the blood request (hospital name, location, contact).
FR-D6	Donor shall view information page about donation rules and eligibility.

HOSPITAL MODULE

IDs	Functional Requirement
FR-H1	Hospital shall register with name, phone, address, and location, license number.
FR-H2	Hospital shall login .
FR-H3	Hospital shall create a blood request (blood type needed, quantity, urgency).
FR-H4	Hospital shall view list of nearby compatible donors with distance.
FR-H5	Hospital shall contact donors directly via the displayed phone number.
FR-H6	Hospital shall monitor and update the blood stock.
FR-H7	Hospital shall approve or reject donors.

ADMIN MODULE

IDs	Functional Requirement
FR-A2	Admin shall login special credentials.
FR-A3	Admin shall approve/reject hospital registrations.
FR-A4	Admin shall monitor donors and hospitals.
FR-A5	Admin shall monitor all blood requests and their status.
FR-A6	Admin shall generate reports (number of donors, requests, blood types availability).
FR-A7	Admin shall manage and reply to messages received from the Contact Us page.

NON-FUNCTIONAL REQUIREMENTS



PERFORMANCE REQUIREMENTS

ID	Requirement
NFR-P1	The system shall respond to user requests within 5 seconds under normal operating conditions.
NFR-P2	The system shall deliver notifications within 30 seconds of blood request creation.

SECURITY REQUIREMENTS

ID	Requirement
NFR-S1	The system shall store all passwords using strong hashing algorithms such as bcrypt or PHP password_hash() .
NFR-S2	The system shall implement role-based access control for Donor, Hospital, and Administrator roles.
NFR-S3	The system shall be protected against common web security attacks , including SQL Injection and Cross-Site Scripting (XSS).

RELIABILITY AND AVAILABILITY

ID	Requirement
NFR-R1	The system shall maintain continuous availability 24 hours per day, 7 days per week.
NFR-R2	The system shall perform automatic database backup on a daily basis.

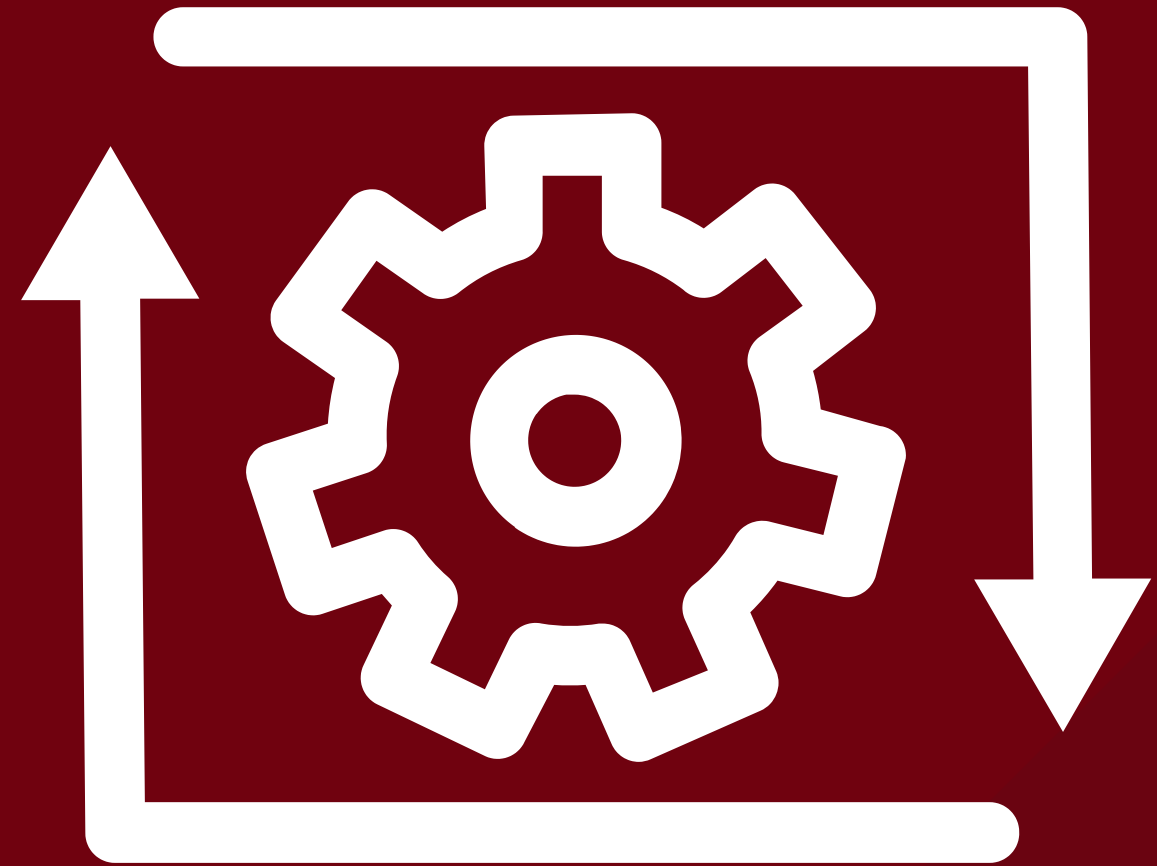
USABILITY

ID	Requirement
NFR-U1	The system interface shall be simple , clear, and easy to use for all users
NFR-U2	The web application shall be fully responsive , providing an optimal viewing and interaction experience across a wide range of devices, including mobile phones, tablets, and desktop computers.
NFR-U3	system shall provide intuitive navigation and consistent user interaction .

SCALABILITY

ID	Requirement
NFR-SC1	The system architecture shall support future expansion , including integration with an iOS application.
NFR-SC2	The system architecture shall support future deployment on cloud infrastructure.

SYSTEM FEATURES



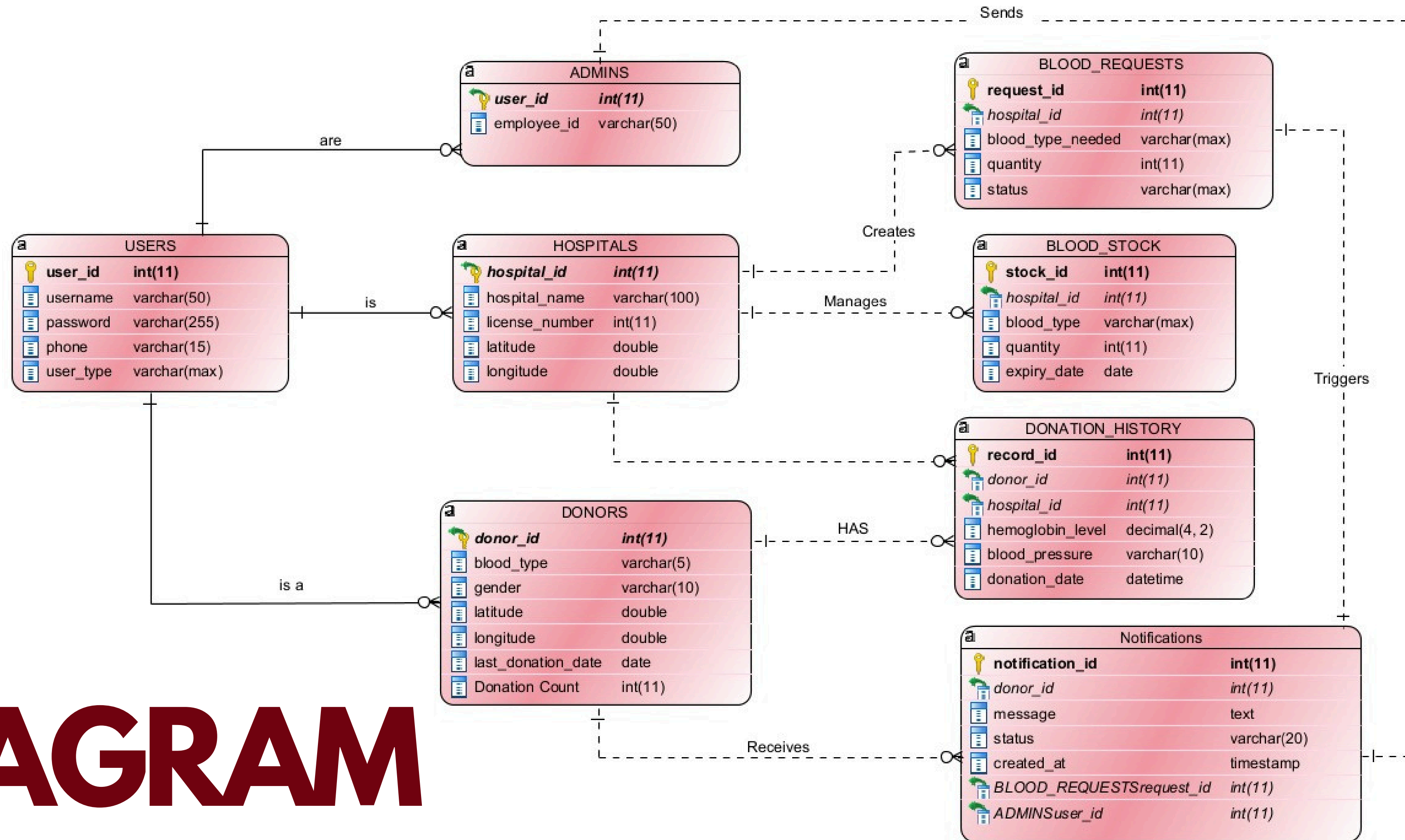
SYSTEM FEATURES

- The system shall **automatically match** blood donation requests with compatible donors based on blood type and geographic location using the Haversine Formula.
- The system shall provide a **real-time notification system for urgent blood requests**.
- The system shall provide a **public About Us page** containing project information, vision, mission, team details, and objectives.
- The system shall provide a public **Contact Us** page with a form that allows donors, hospitals, and visitors to send messages or inquiries directly to the **administration**.
- The system shall **mask password** fields during user registration and login processes.
- The system shall provide a **responsive web design compatible with desktops, tablets, and mobile devices**.

SYSTEM FEATURES

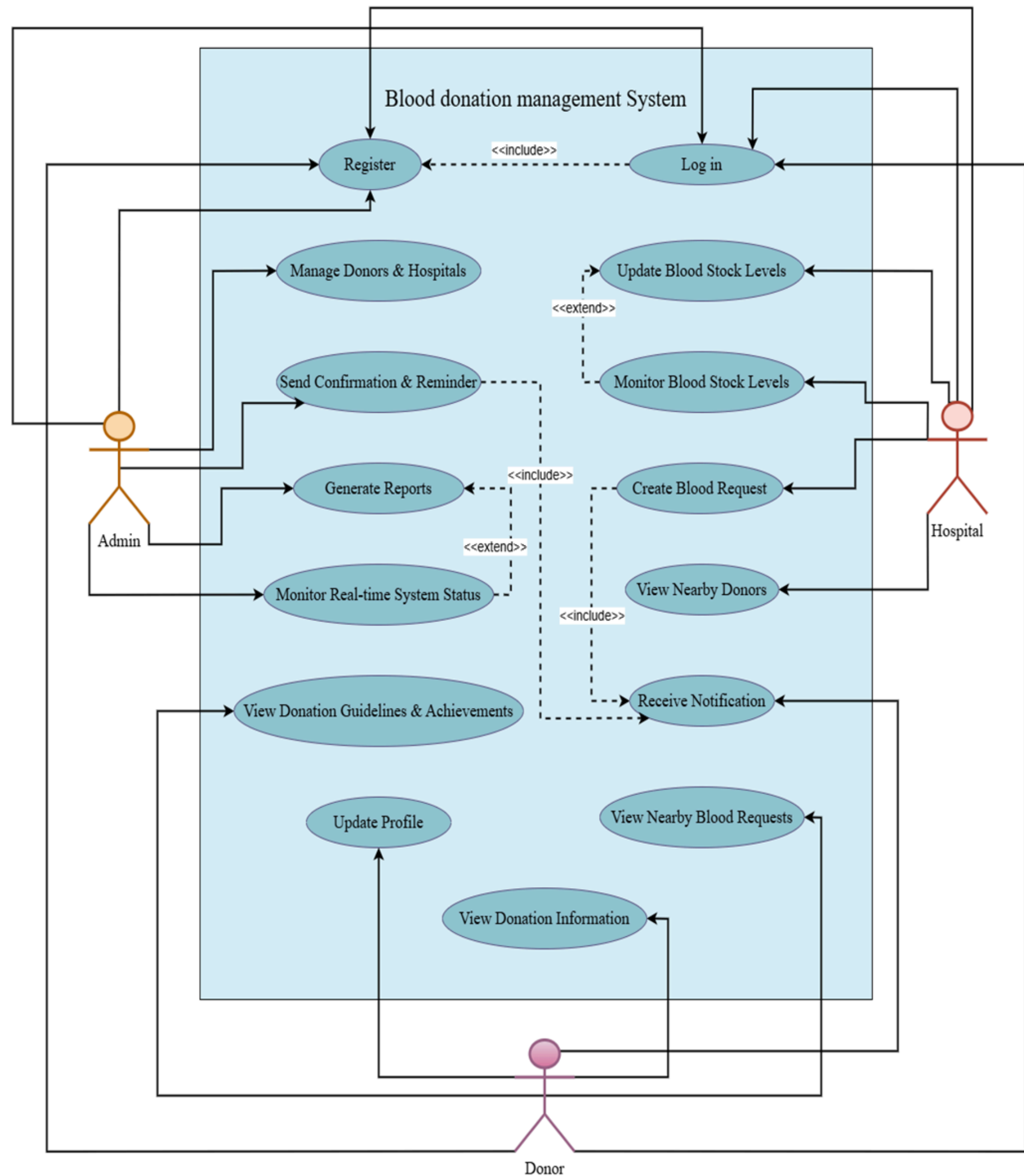
- The system shall send **blood donation requests simultaneously to multiple compatible donors** based on blood type, location, and availability status.
- The system shall **continue searching for compatible donors until the hospital confirms** that the blood request has been fulfilled.
- The system shall continue **sending notifications** to other compatible donors even if one donor accepts the request, until the request status is updated to **“Fulfilled.”**
- The system shall automatically send a **confirmation notification** to the donor after the donation is marked as completed by the hospital or administrator.
- The system shall automatically send a **reminder** notification to the donor 3 months after their last donation date to encourage future donations.

ER DIAGRAM

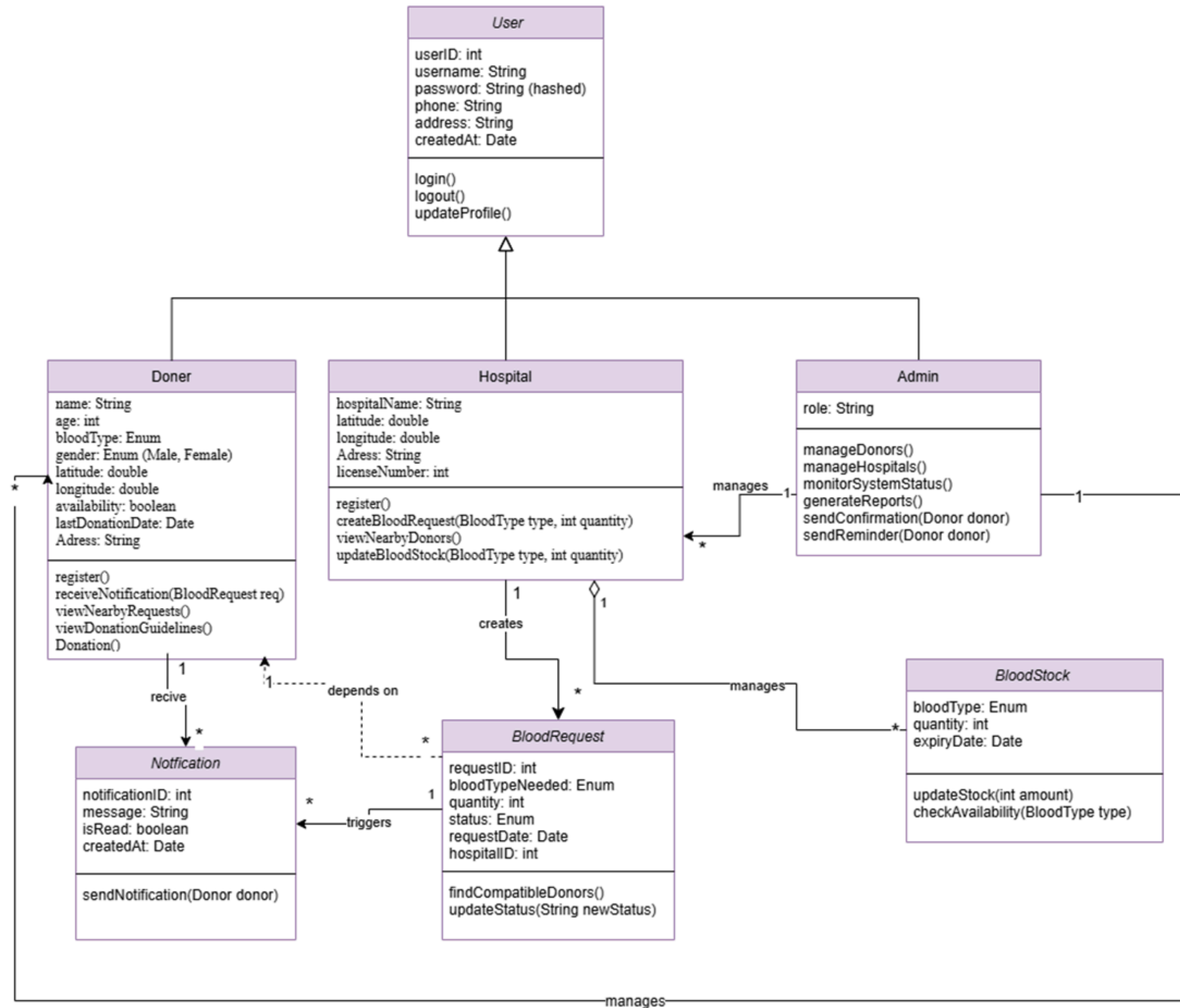


UML DIAGRAMS

1. USE CASE DIAGRAM

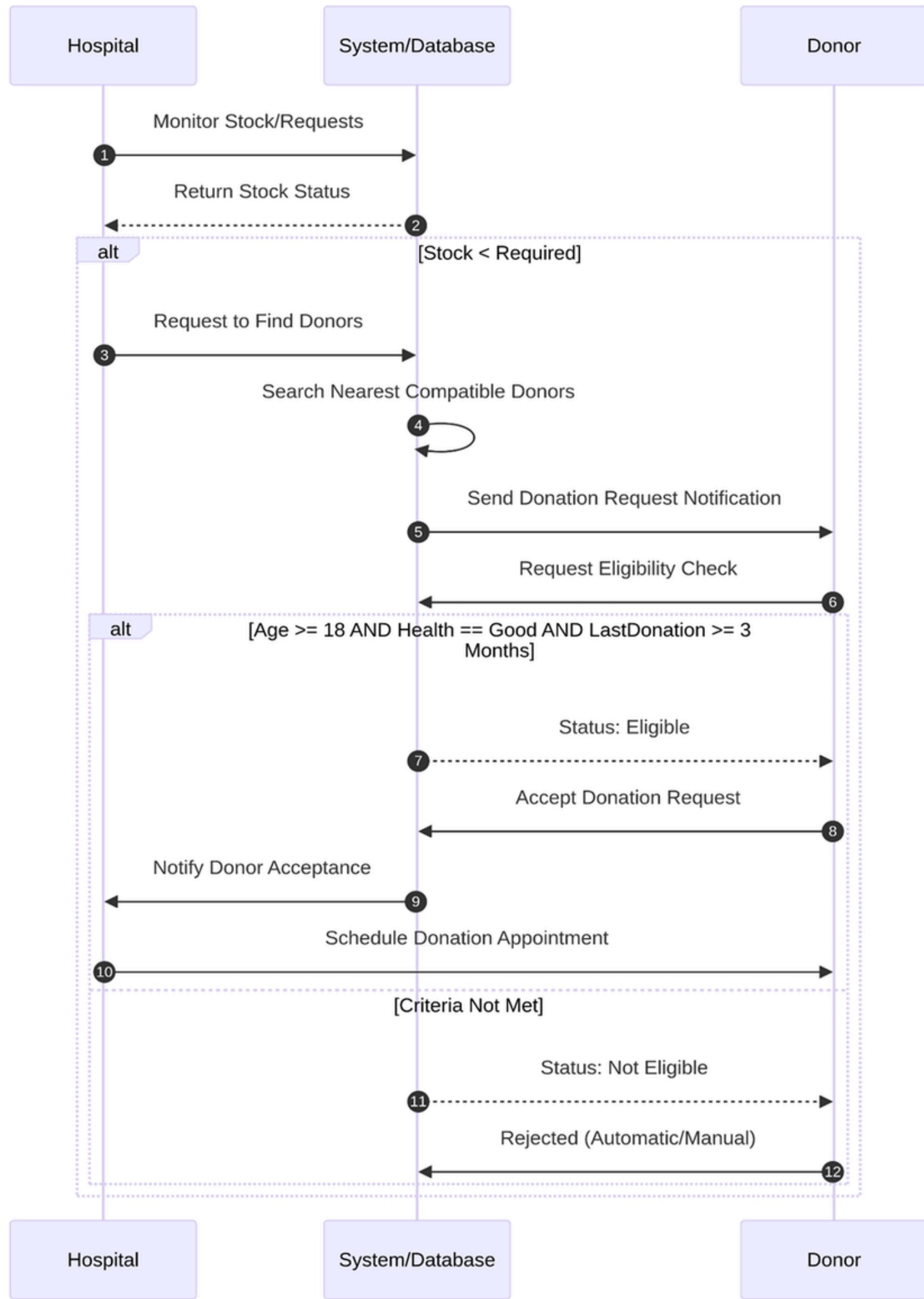


2. CLASS DIAGRAM

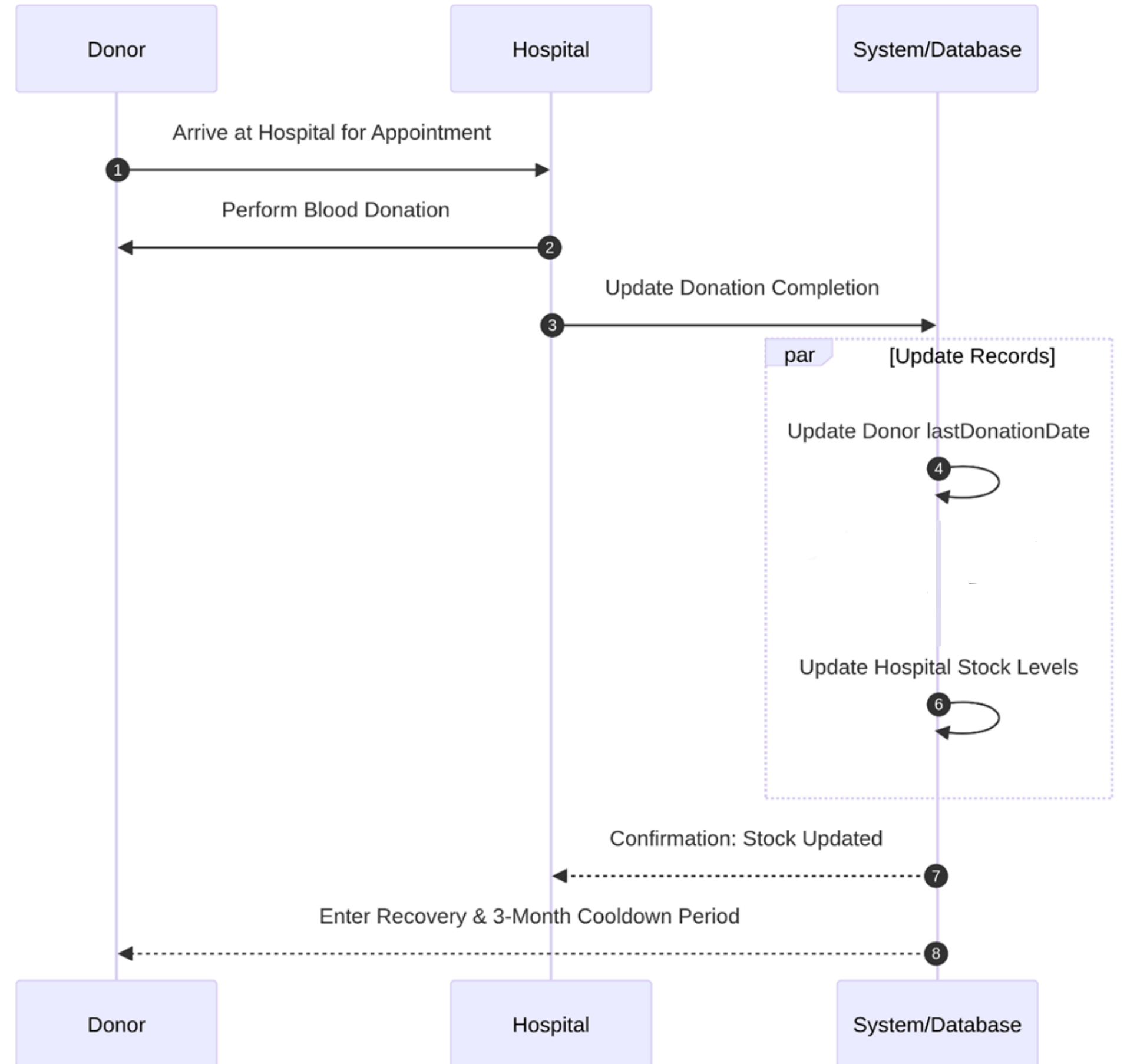


3 . SEQUENCE DIAGRAM

1. Blood Request and Donor Notification Scenario



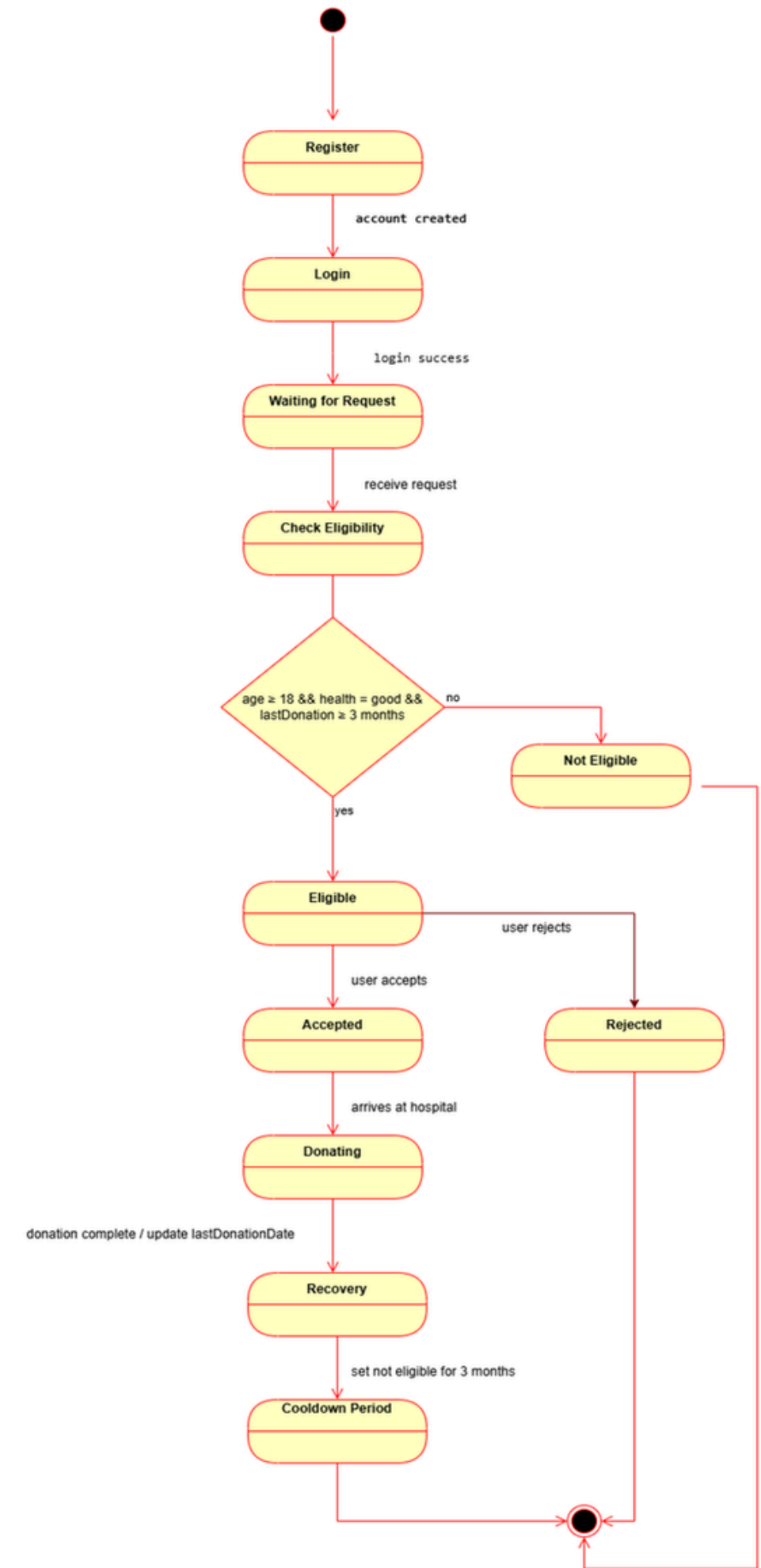
2. Donation Process and Stock Update Scenario



4. STATE MACHINE DIAGRAMS

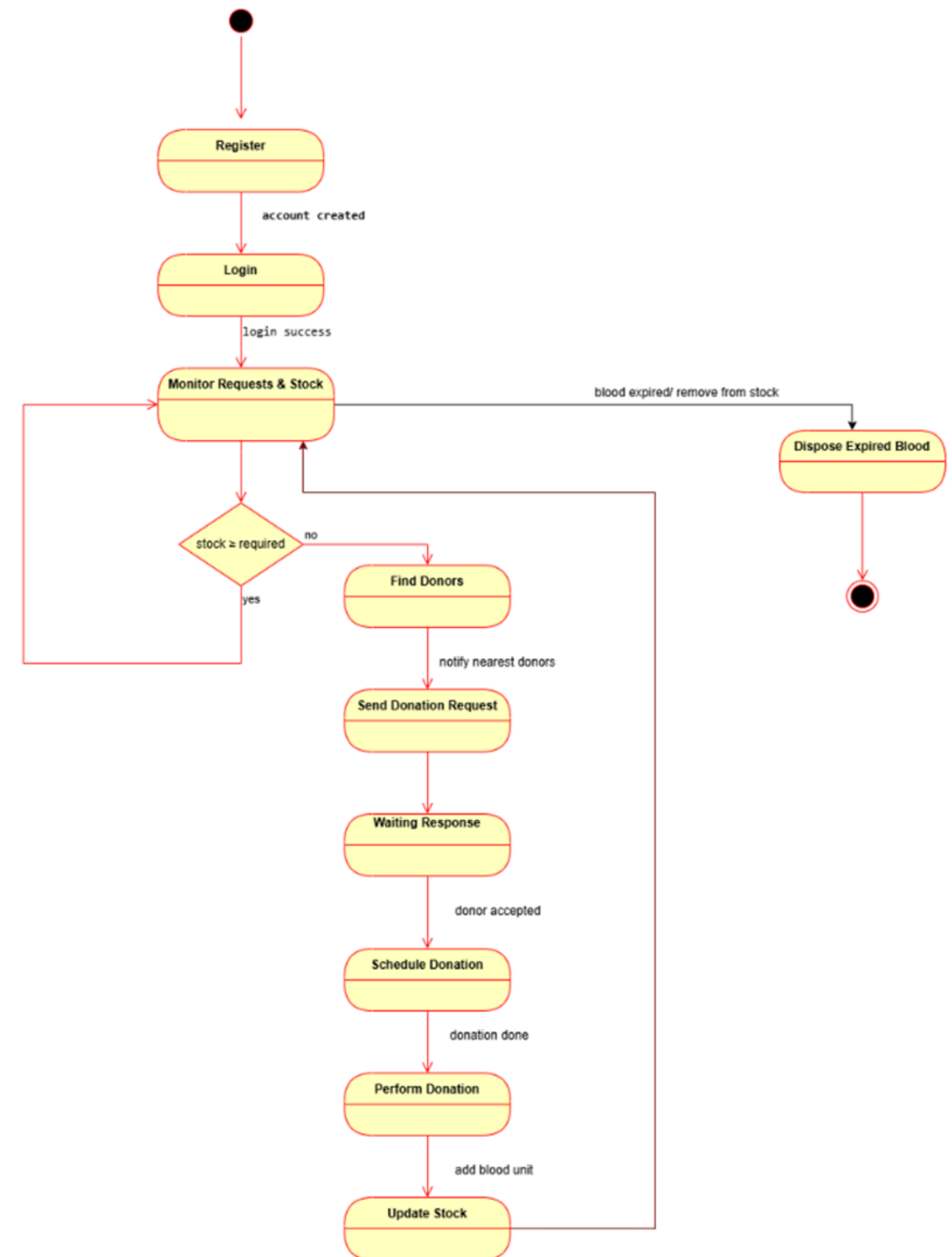
DONOR STATE MACHINE

Donor State Machine

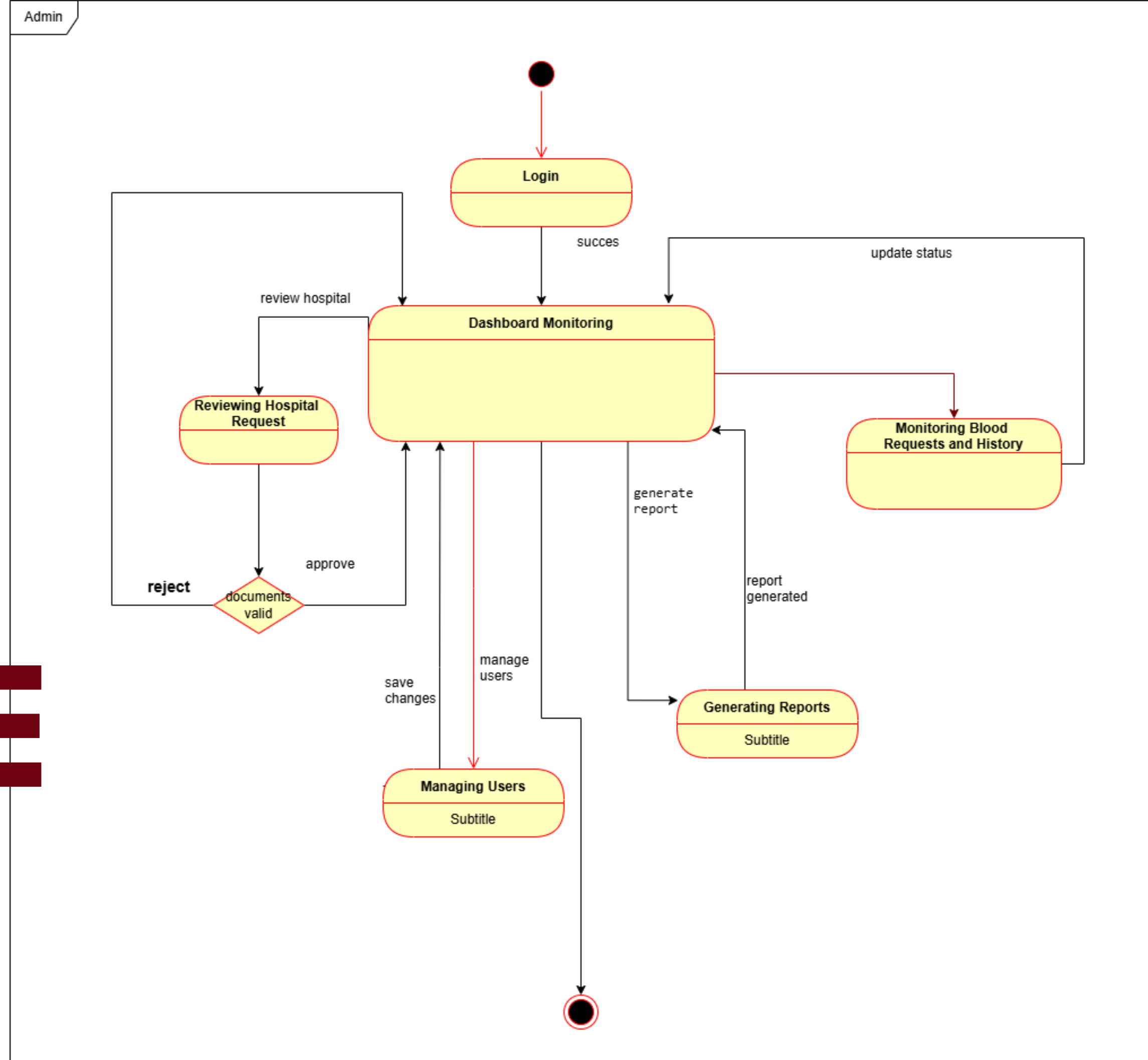


HOSPITAL STATE MACHINE

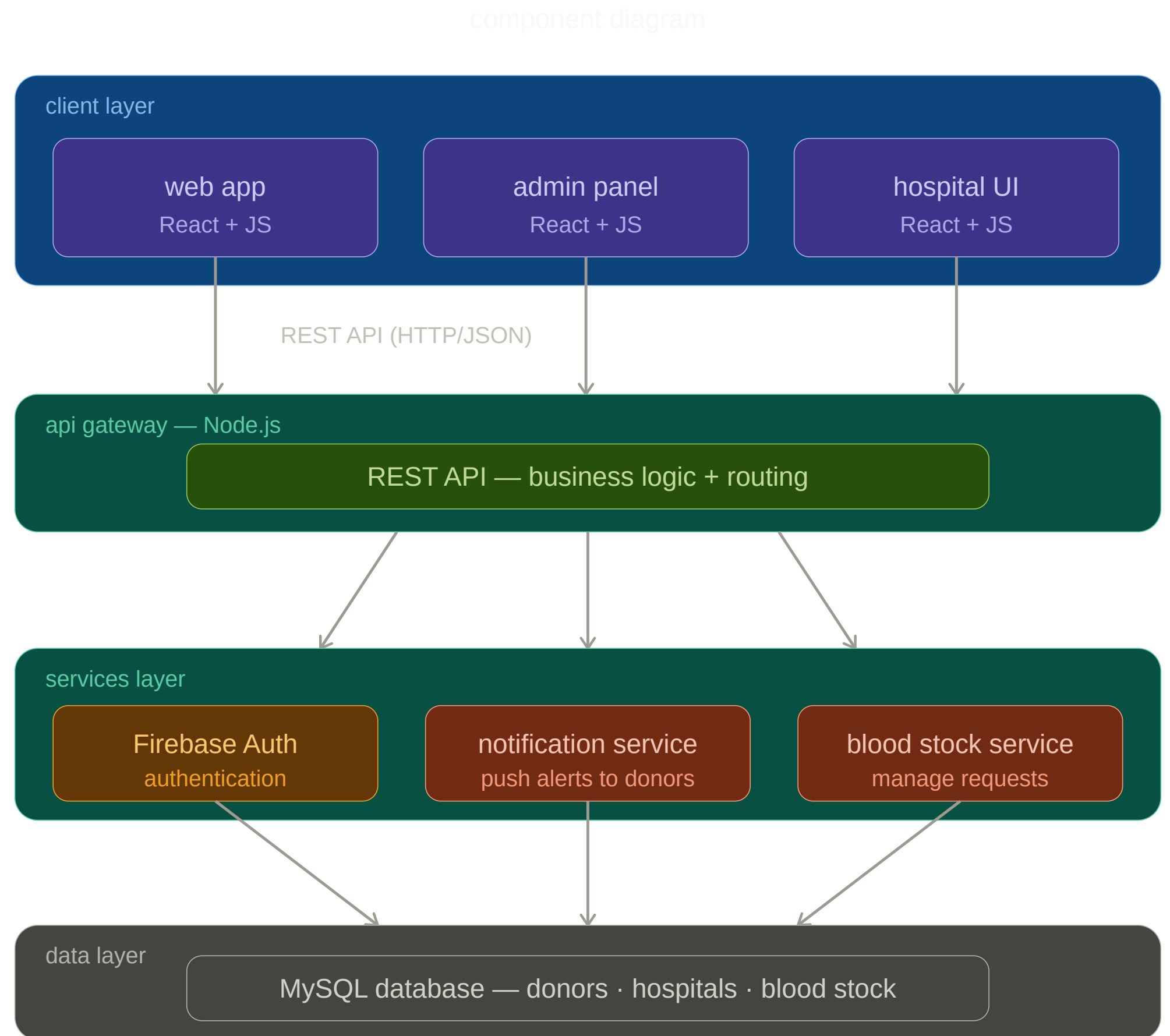
Hospital State Machine



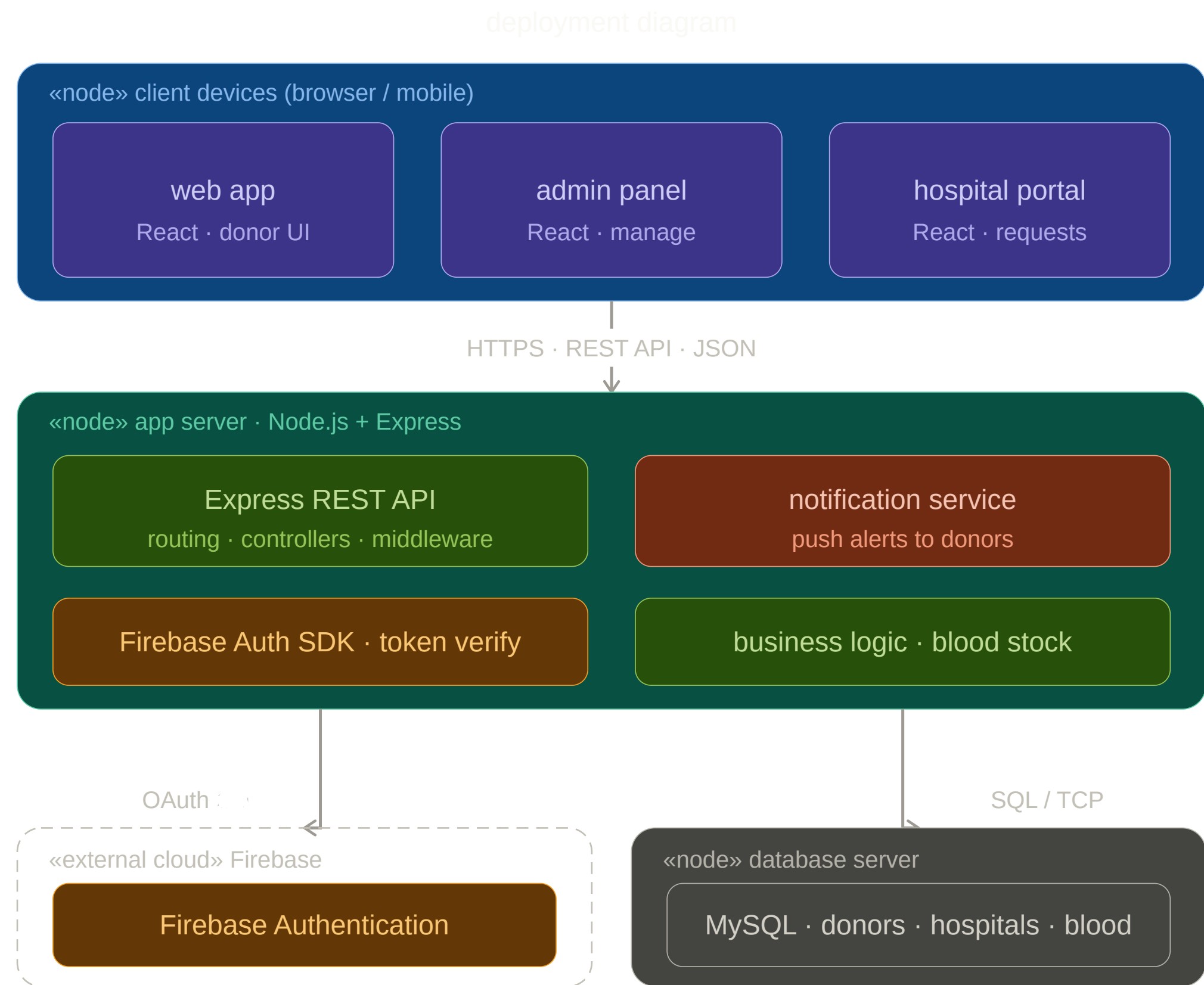
ADMIN STATE MACHINE



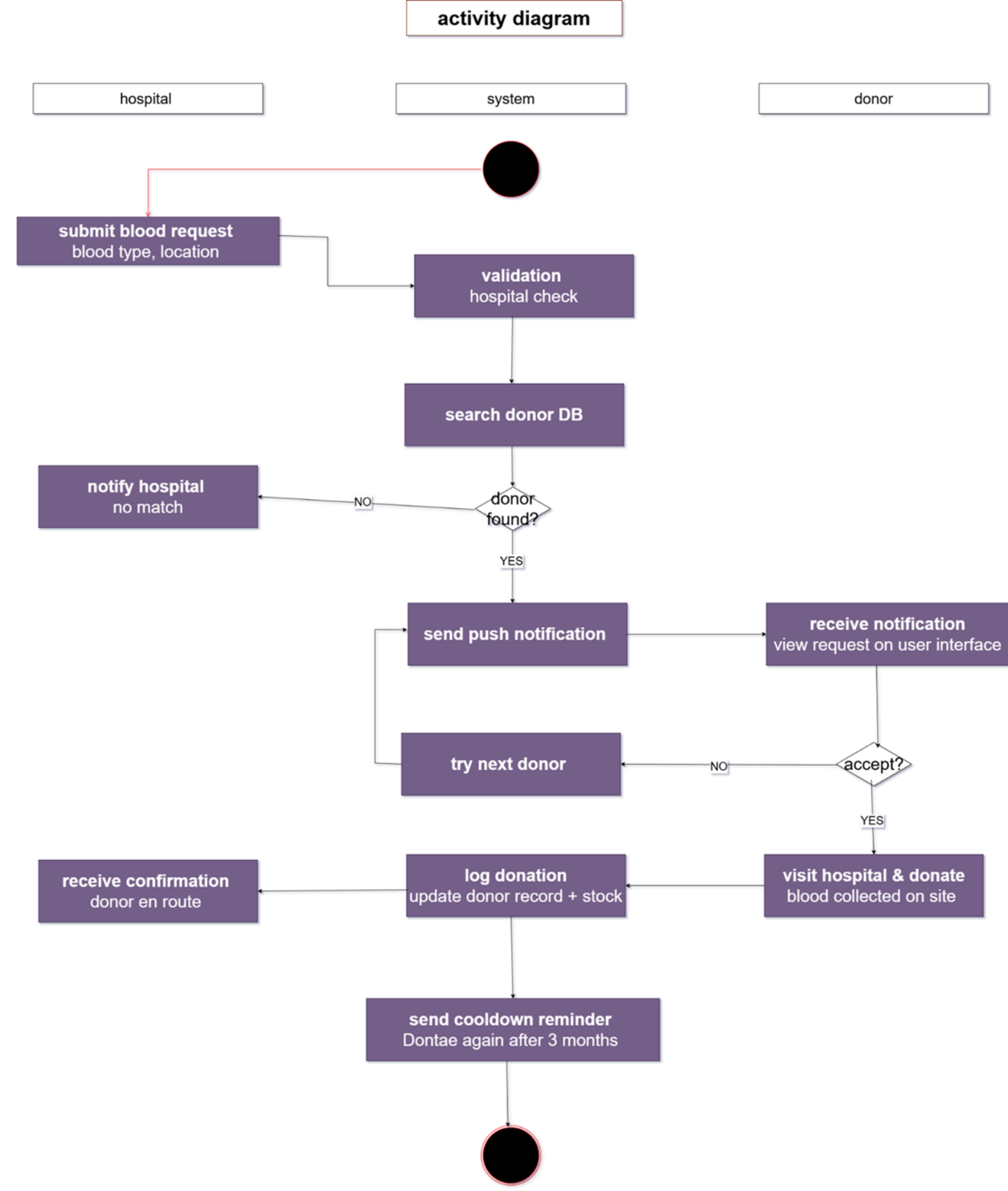
5. COMPONENT DIAGRAM



6 . DEPLOYMENT DIAGRAM



7. ACTIVITY DIAGRAM



REQUIREMENTS TRACEABILITY MATRIX (RTM)

DONOR MODULE

Req ID	Requirement Description	Test Case ID	Test Scenario & Expected Result	Test Status
FR-D1	Donor shall be able to register with name, age, phone, address, blood type, gender, and location.	TC-D01	Verify successful registration when a donor inputs all valid required data.	
FR-D2	Donor shall login using username and password.	TC-D02	Verify successful login with correct credentials and rejection of invalid ones.	
FR-D3	Donor shall Viwe his/her profile (including availability and last donation date).	TC-D03	Verify donor can successfully modify profile details and save changes.	
FR-D4	Donor shall receive push notification when a nearby hospital needs his/her blood type.	TC-D04	Verify FCM triggers a push notification to the donor's device upon a matching request.	
FR-D5	Donor shall view details of the blood request (hospital name, location, contact).	TC-D05	Verify notification opens a detailed view displaying accurate hospital request data.	
FR-D6	Donor shall view information page about donation rules and eligibility.	TC-D06	Verify the donation rules/eligibility page loads correctly and is readable.	

HOSPITAL MODULE

Req ID	Requirement Description	Test Case ID	Test Scenario & Expected Result	Test Status
FR-H1	Hospital shall register with name, phone, address, and location.	TC-H01	Verify hospital account is created successfully pending admin approval.	
FR-H2	Hospital shall login.	TC-H02	Verify authorized hospital personnel can log in successfully.	
FR-H3	Hospital shall create a blood request (blood type needed, quantity, urgency).	TC-H03	Verify system accepts and broadcasts a new blood request with proper parameters.	
FR-H4	Hospital shall contact donors directly via the displayed phone number.	TC-H04	Verify donor contact details are accurately displayed to the hospital user.	
FR-H5	Hospital shall accept or reject incoming donor requests based on medical eligibility.	TC-H05	Verify that the hospital can approve a donor request or reject it with a medical reason, and that the donor status is updated accordingly.	

ADMINS MODULE

Req ID	Requirement Description	Test Case ID	Test Scenario & Expected Result	Test Status
FR-A1	Admin shall login with special credentials.	TC-A01	Verify admin dashboard access is strictly granted to admin credentials.	
FR-A2	Admin shall approve/reject hospital registrations.	TC-A02	Verify admin can successfully change hospital status to 'Approved' or 'Rejected'.	
FR-A3	Admin shall monitor donors or hospitals.	TC-A03	Verify admin possesses full CRUD capabilities for users.	
FR-A4	Admin shall monitor all blood requests and their status.	TC-A04	Verify admin dashboard displays a real-time, accurate list of all active/past requests.	
FR-A5	Admin shall generate reports (number of donors, requests, blood types availability).	TC-A05	Verify admin can export accurate statistical reports based on system data.	

SYSTEM FEATURES

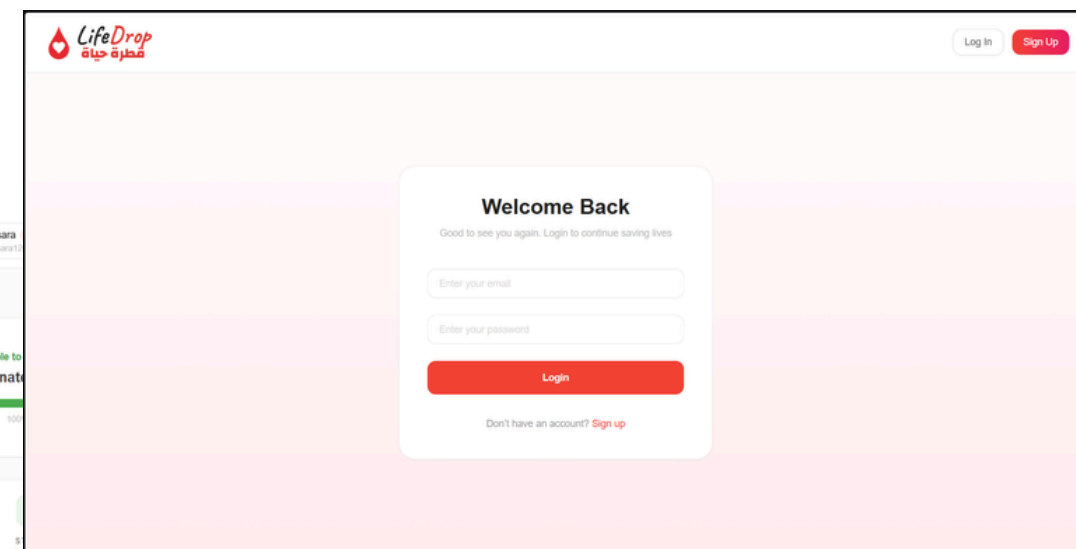
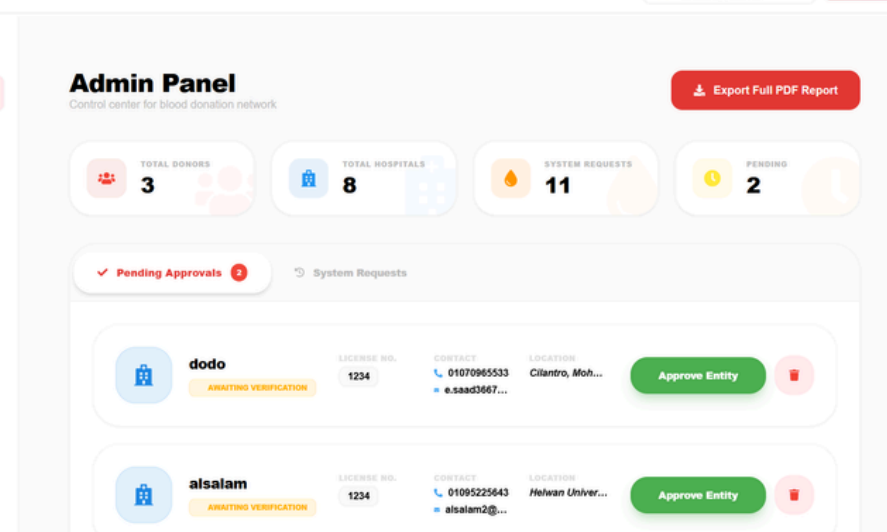
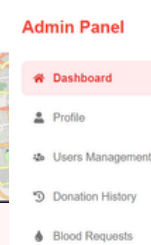
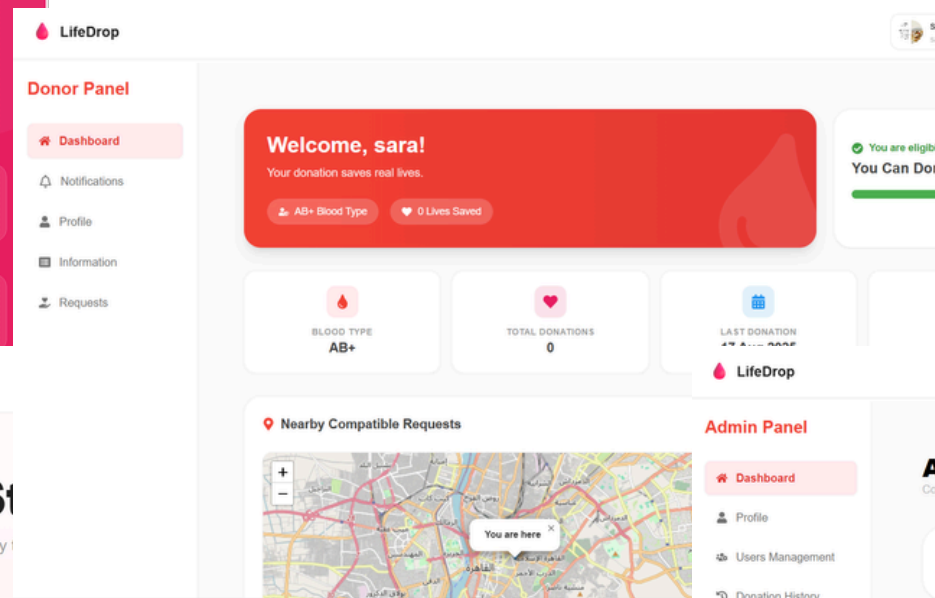
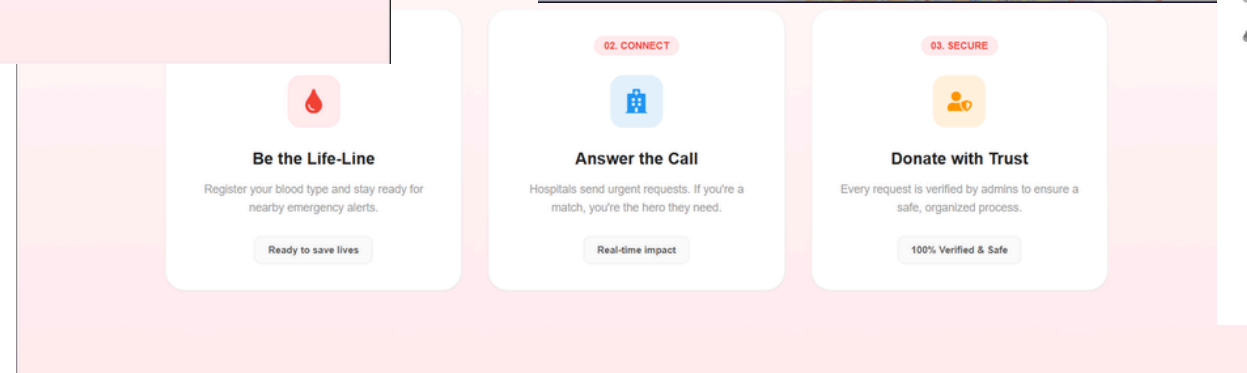
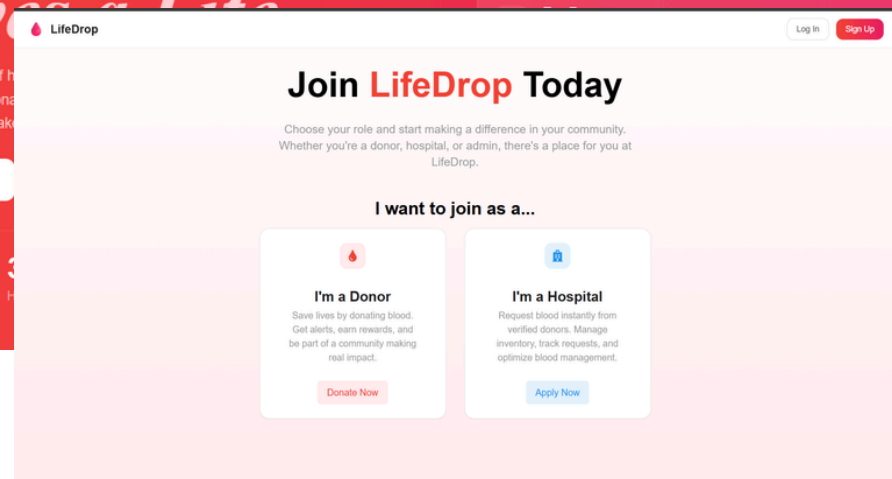
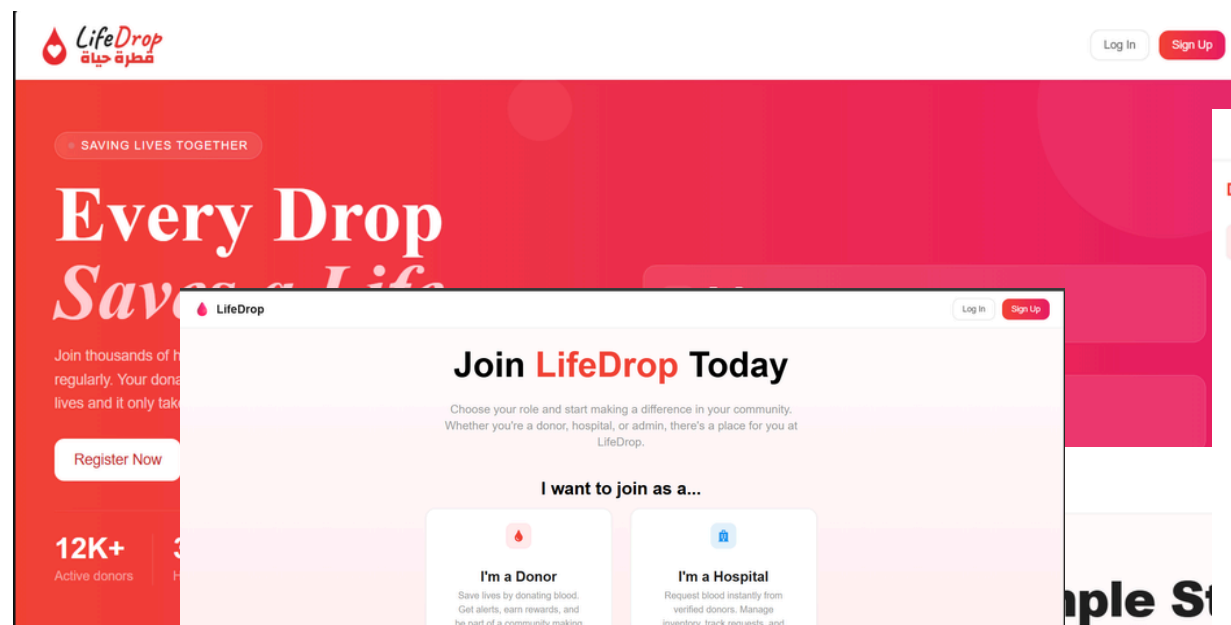
Req ID	Requirement Description	Test Case ID	Test Scenario & Expected Result	Test Status
SYS-01	Smart Compatibility Engine: match donors and recipients beyond identical types.	TC-S01	Verify system suggests O- donors for A- requests based on the compatibility matrix.	
SYS-02	Automated Deferral Logic: enforce 3-month waiting period and medical rejection.	TC-S02	Verify system blocks donation requests for users who donated within the last 90 days.	
SYS-03	Automatic matching of blood type and location (using Haversine formula).	TC-S03	Verify the calculated distance between hospital and donor coordinates is accurate.	
SYS-04	Security: Password not visible during registration (masked input).	TC-S04	Verify that the password field input is masked (e.g., asterisks) during registration.	

NON-FUNCTIONAL REQUIREMENTS

Req ID	Requirement Description	Test Case ID	Test Scenario & Expected Result	Test Status
NFR-P1	System shall respond to requests within 5 seconds.	TC-N01	Verify API response times remain under 5 seconds during load testing.	
NFR-P2	Notification shall be delivered within 30 seconds of request creation.	TC-N02	Verify FCM latency between hospital request submission and donor receipt is	
NFR-S1	All passwords shall be stored using strong hashing (bcrypt/PHP password_hash).	TC-N03	Verify database inspection shows passwords stored purely as bcrypt hashes.	
NFR-S2	Role-based access control (Donor / Hospital / Admin).	TC-N04	Verify access control restricts endpoints appropriately based on user role.	
NFR-S3	Protection against SQL injection and XSS.	TC-N05	Verify input fields sanitize malicious scripts and SQL queries during testing.	
NFR-S4	Medical Data Privacy: All clinical screening results and health history	TC-N06	Verify HIPAA-compliant encryption standards are applied to medical history	
NFR-R1	System shall be available 24/7.	TC-N07	Verify server uptime and redundancy configurations.	
NFR-R2	Database backup shall be performed daily.	TC-N08	Verify automated daily cron jobs execute and store SQL database dumps securely.	
NFR-U1	Interface shall be simple/intuitive for all age groups.	TC-N09	Verify UI/UX passes user acceptance testing (UAT).	



Let's go through the website



THANK YOU

﴿وَمَنْ أَحْيَاهَا فَكَأَنَّمَا أَحْيَا النَّاسَ جَمِيعًا﴾ [سورة المائدة: 32]